

Feasibility Limits of De-Extinction: A Meta-Analysis of Evidence Availability and Biological Constraints

Input Question

Is de-extinction possible?

Domain

Biology

Validation Status

needs_data

Problem Statement

The feasibility of de-extinction is constrained by the physical limits of DNA survival over geological timescales and the technical capabilities of current genome editing tools like CRISPR. Specifically, it is unclear which extinct species can be resurrected given these constraints, and whether the resulting organisms would be true clones or hybrid entities.

Rationale

Understanding the biological boundaries of de-extinction requires distinguishing between species with recoverable genetic material (e.g., recent extinctions like the mammoth) and those with degraded DNA (e.g., dinosaurs). The provided evidence corpus is insufficient to support specific biological claims, necessitating a research plan that explicitly identifies these knowledge gaps and proposes a framework for verifying feasibility through domain-specific literature and experimental design.

Generated Hypotheses

Hypothesis 1

- **Hypothesis**: INSUFFICIENT_EVIDENCE: No valid scientific hypothesis regarding the feasibility of de-extinction can be generated because the allowed evidence corpus (EV-Q034-5445f89d58a728a64911f1ac, EV-Q034-82b7ab3947eb1574651216fb, EV-Q034-39aef2c6325100af5d9f7dff) contains no domain-relevant data on CRISPR, ancient DNA survival limits, or de-extinction ethics.
- **Mechanism**: The provided evidence cards address unrelated domains: COVID-19 forecasting (EV-Q034-5445f89d58a728a64911f1ac), consciousness oscillators (EV-Q034-82b7ab3947eb1574651216fb), and neutrino physics (EV-Q034-39aef2c6325100af5d9f7dff). Without biological or genetic evidence, no mechanistic chain linking genome editing technologies to species resurrection can be constructed or validated within system constraints.

- ****Falsifiable Prediction****: If any of the allowed evidence IDs are found to contain verified content regarding 'de-extinction', 'CRISPR-mediated genome editing in extinct species', or 'ancient DNA degradation kinetics', this declaration of insufficient evidence is falsified.
- ****Required Observations****: Verification that EV-Q034-5445f89d58a728a64911f1ac relates exclusively to pandemic forecasting and not genomics ; Verification that EV-Q034-82b7ab3947eb1574651216fb relates exclusively to neural oscillators and not paleogenetics ; Verification that EV-Q034-39aef2c6325100af5d9f7dff relates exclusively to particle physics and not biology ; Confirmation that knowledge gaps for 'CRISPR feasibility', 'ancient DNA survival limits', and 'de-extinction ethics' remain unfilled by allowed sources
- ****Risk of Being Wrong****: Low risk of factual error regarding the content of allowed evidence; high risk of user dissatisfaction if external knowledge was implicitly expected despite strict adherence to allowed_evidence_ids constraints.

Technical Details

This experiment design addresses the 'insufficient_evidence' state identified in the recommended hypothesis. The primary objective is to formally verify that the allowed evidence corpus (EV-Q034-5445f89d58a728a64911f1ac, EV-Q034-82b7ab3947eb1574651216fb, EV-Q034-39aef2c6325100af5d9f7dff) contains no domain-relevant information regarding de-extinction, CRISPR, or ancient DNA. The technical approach involves a systematic content audit using keyword matching and semantic similarity analysis against a controlled vocabulary of de-extinction biology. This meta-scientific verification ensures strict adherence to evidence grounding constraints by explicitly documenting the absence of relevant data, rather than attempting to construct a biological mechanism from unrelated domains (epidemiology, neuroscience, particle physics).

Datasets

Source

```
```json
[
{
 "id": "EV-Q034-5445f89d58a728a64911f1ac",
 "description": "ArXiv paper on COVID-19 forecasting models; verified as irrelevant to genetics/de-extinction."
},
{
 "id": "EV-Q034-82b7ab3947eb1574651216fb",
 "description": "ArXiv paper on systems of oscillators and consciousness; verified as irrelevant to genetics/de-extinction."
},
{

```

```
"id": "EV-Q034-39aef2c6325100af5d9f7dff",
"description": "ArXiv paper on neutrino physics; verified as irrelevant to genetics/de-extinction."
}
]
...
```

## Target

```
```json  
{  
  "description": "Structured audit report confirming the absence of de-extinction related content in the source  
  datasets, validating the 'insufficient_evidence' status.",  
  "format": "JSON Audit Log"  
}  
...
```

Paper Abstract

Background: De-extinction efforts rely on the preservation of ancient DNA and the precision of genome editing tools like CRISPR. However, the viability of such projects varies significantly across species due to DNA degradation rates. Methods: This study conducts a systematic audit of available evidence to determine the current state of knowledge regarding de-extinction feasibility. We analyze the relevance of existing literature to key biological mechanisms, including DNA survival limits and hybridization techniques. Validation Plan: We propose a validation protocol involving keyword scanning and semantic similarity analysis of relevant biological databases to identify gaps in current evidence. Results: Pending execution of verification experiments. Current analysis indicates a critical lack of domain-specific evidence in the provided corpus, highlighting the need for targeted data collection in paleogenomics and CRISPR applications.

Methods

1. Keyword Definition: Establish a strict query set $Q = \{'CRISPR', 'Cas9', 'de-extinction', 'ancient DNA', 'genome editing', 'mammoth', 'cloning', 'paleogenomics', 'species resurrection'\}$.
2. Content Scanning: Perform full-text search of the three allowed EvidenceCards for any occurrence of terms in Q .
3. Semantic Verification: Calculate cosine similarity between the text embeddings of the EvidenceCards and a reference vector representing de-extinction literature (used only as a negative control query).
4. Relevance Thresholding: Set a relevance threshold of 0.0. Any score above this would trigger a manual review.

Experiments

Baselines

```
```json
```

```
[
```

```
"Random Chance Baseline: Expectation of finding relevant biological terms in random physics/neuroscience papers is negligible.",
```

```
"Keyword Match Baseline: Direct string matching for 'DNA' or 'gene' which may appear metaphorically or in unrelated contexts but not in the context of genome editing for de-extinction."
```

```
]
```

```
```
```

Metrics

```
```json
```

```
[
```

```
"Term Frequency-Inverse Document Frequency (TF-IDF) score for target keywords.",
```

```
"Semantic Similarity Score (Cosine Similarity) against de-extinction query vectors.",
```

```
"Binary Relevance Flag (0 for irrelevant, 1 for relevant)."
```

```
]
```

```
```
```

Ablation

Remove the constraint of 'allowed_evidence_ids' and allow external search to demonstrate the contrast between the current knowledge gap and the actual state of the field (counterfactual analysis only).

Validation Protocol

Cross-validate the audit results by having two independent NLP pipelines (one rule-based, one embedding-based) analyze the same documents. Consensus on 'irrelevance' confirms the hypothesis.

Results

当前状态：待执行验证实验。系统已生成可复现实验脚本、数据字段清单与评价指标，尚未运行真实实验。

References

- **EV-Q034-5445f89d58a728a64911f1ac** · arxiv · arXiv:2007.02105
- authors: Not available · year: Not available

- url: <https://arxiv.org/pdf/2007.02105.pdf> · doi: Not available
- reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:2|section:page-2|paragraph:1;
content_sha256=03f8aa647cd75f5ab874d9c89ad9249bdaa8838251f32e35df3f1f877656958c
- **EV-Q034-82b7ab3947eb1574651216fb** · arxiv · arXiv:1903.02594
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/1903.02594.pdf> · doi: Not available
- reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:1|section:page-1|paragraph:1;
content_sha256=0c2313ea416677a80094859716addca0a26ad00d6c03ec5c2c9964c0f8dc507a
- **EV-Q034-39aef2c6325100af5d9f7dff** · arxiv · arXiv:hep-ph/9705325
- authors: Not available · year: Not available
- url: <https://arxiv.org/pdf/hep-ph/9705325.pdf> · doi: Not available
- reliability_note: eligibility_status=FULLTEXT_VERIFIED;
topic_relevance_status=DIRECT_QUESTION_CORE; locator=page:2|section:page-2|paragraph:1;
content_sha256=136034761e52c5cd5babe265f8efa85d3dfec165bb0626387dc98fe0176fc0b7

Reviewer Comments

- The revised plan correctly implements the 'INSUFFICIENT_EVIDENCE' declaration as the primary hypothesis, strictly adhering to system constraints regarding the absence of domain-relevant evidence.
- All scoring metrics (novelty, falsifiability, feasibility, evidence_support) are correctly set to 0.0, eliminating any implication of partial validity for an untestable state.
- Irrelevant Hypothesis 1 (linking pandemic forecasting to de-extinction) has been successfully removed, resolving the critical issue of cross-domain noise.
- Knowledge gaps are explicitly defined with empty evidence_ids arrays, accurately reflecting the lack of supporting data in the allowed corpus.
- Experiment design is appropriately framed as a meta-scientific audit to verify the absence of evidence, with valid baselines and metrics for this specific verification task.
- Results field remains correctly marked as pending/unexecuted, avoiding fabrication.

Revision History

- auto_revision_1: 依据评审意见重做假设与实验设计。

Reproducibility Checklist

- Verify that the list of allowed_evidence_ids is strictly limited to the three provided IDs.

- Ensure the keyword list Q is comprehensive for de-extinction biology.
- Confirm that no external biological databases were queried during the primary audit.
- Document the specific sections of each EvidenceCard scanned (e.g., abstract, introduction, conclusion).